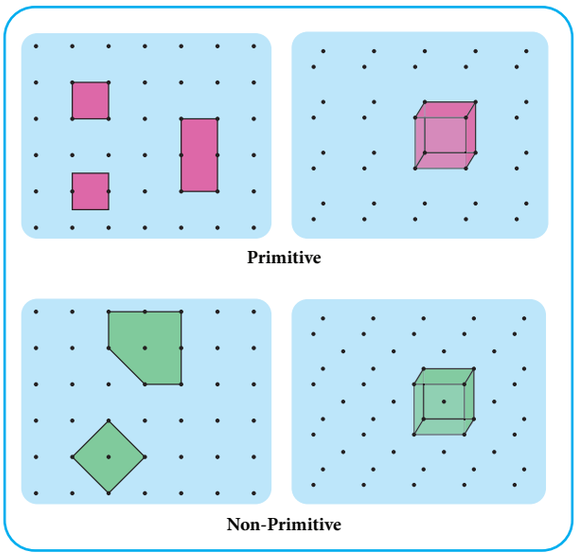
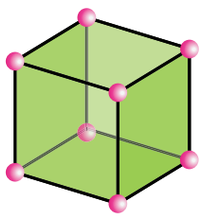
6.5 Primitive and non-primitive unit cell

There are two types of unit cells: primitive and non-primitive. A unit cell that contains only one type of lattice point is called a primitive unit cell, which is made up from the lattice points at each of the corners.

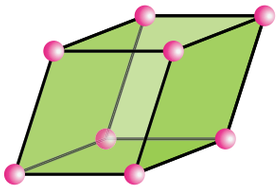
In case of non-primitive unit cells, there are additional lattice points, either on a face of the unit cell or with in the unit cell.



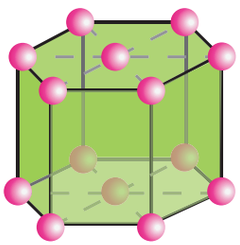
There are seven primitive crystal systems; cubic, tetragonal, orthorhombic, hexagonal, monoclinic, triclinic and rhombohedral. They differ in the arrangement of their crystallographic axes and angles. Corresponding to the above seven, Bravais defined 14 possible crystal systems as shown in the figure.



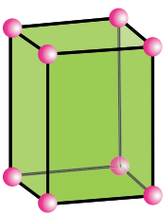
Cubic
$a = b = c$
$\alpha = \beta = \gamma = 90°$



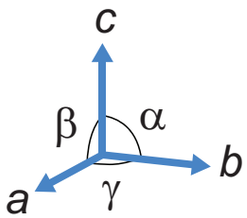
Rhombohedral
$a = b = c$
$\alpha = \beta = \gamma \neq 90°$

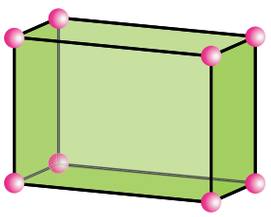


Hexagonal
$a = b \neq c$
$\alpha = \beta = 90°, \gamma = 120°$

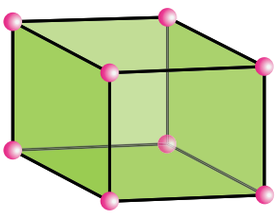


Tetragonal
$a = b \neq c$
$\alpha = \beta = \gamma = 90°$

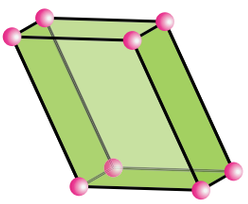




Orthorhombic
$a \neq b \neq c$
$\alpha = \beta = \gamma = 90°$



Monoclinic
$a \neq b \neq c$
$\alpha = \gamma = 90°, \beta \neq 90°$



Triclinic
$a \neq b \neq c$
$\alpha \neq \beta \neq \gamma \neq 90°$

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